

UNDERSTANDING script.js

Student Score Calculator / Code guide

A plain-language explanation of how the page validates scores and calculates student totals and grades.

What the script does

The script connects the score-entry table to live totals, letter grades, and input validation. It runs in the browser and responds whenever a score field changes.

In one sentence: validate the edited score, then recalculate that student's result when all four scores are valid and present.

How the page and script connect

The HTML loads script.js with the defer attribute, so the browser parses the page before running the script.

The script selects inputs marked data-student and the validation message element. For results, it finds the .total and .grade cells in the student's table row.

FLOW: User edits score > Check input > Find that student's fields > Show total and grade, or dashes

Walkthrough

1. Find page elements. `querySelectorAll` collects all student score inputs. `querySelector` finds the validation message area.
2. Convert total to grade. `getGrade` checks thresholds from highest to lowest: 70 or more is A; 60 B; 50 C; 45 D; 40 E; anything lower is F. Checking in descending order lets the first matching threshold decide.
3. Recalculate one student. `updateStudent` selects fields for one student, finds their row, and gets the total and grade cells.

If any field is invalid, the result is cleared to an em dash and the grade marker is removed. If any field is blank, it also leaves the result blank. This avoids showing a partial total as final.

When all four values are present and valid, the script converts them to numbers, adds them, displays the total and grade, and stores the grade in data-grade for styling or other page behavior.

Validation and events

Each score field listens for the input event, so the code runs as the user types or changes a value.

A valid value is blank temporarily, or a whole number from zero up to the field's max. CA scores allow 0-10; the exam allows 0-70.

The script records validity in `aria-invalid`. Invalid input displays a message such as â Enter a whole number from 0 to 10 for this score.â When no field is invalid, the message is cleared. The script then recalculates only the student whose field changed.

Grade scale

Total score	Grade
70-100	A
60-69	B
50-59	C
45-49	D
40-44	E
0-39	F

The three continuous-assessment fields are each out of 10 and the exam is out of 70, making 100 the maximum total.

Worked example

Scores: CA1 8, CA2 7, CA3 9, Exam 61. The total is $8 + 7 + 9 + 61 = 85$, so the grade is A.

If CA2 is changed to 12, validation fails because the maximum is 10. The total and grade return to dashes until the score is corrected.

JavaScript ideas used

DOM selection: finding HTML elements in the document.

Events: running code in response to a user action.

Data attributes: data-student associates inputs with a student.

Arrays: spread syntax turns the input collection into an array; some checks conditions, map converts values, and reduce adds them.

Early returns: exit the update function when a result should remain blank.

Accessibility: aria-invalid marks invalid fields, and the live message area announces feedback.

Quick summary

The input event validates one edited field, updates the page's validation message, and recalculates the matching student. Results appear only when all four scores are present and valid.

Based on script.js and its related markup in index.html.